

Integrating Structure Tensor and Dynamic Cognitive Diagnosis for Medical Image Segmentation

Jiajie Wang^{a,1}, Erdong Yuan^{a,1}, Zhaozhao Su^a, Gao Huang^b, Yuxuan Luo^a, Yining Yang^c, Zitong Guo^c, Jing Tao^c, Liejun Wang^{a,*}

^a*School of Computer Science and Technology, Xinjiang University, Urumqi, 830000, Xinjiang, China*

^b*Department of Automation, Tsinghua University, Beijing, 100084, China*

^c*Xinjiang Key Laboratory of Cardiovascular Homeostasis and Regeneration Research, People's Hospital of Xinjiang Uygur Autonomous Region, Urumqi, 830000, Xinjiang, China*

Abstract

Precise boundary delineation and lesion localisation are essential in medical image segmentation. Nevertheless, lesions often coexist with intricate anatomical formations. This complicates the differentiation of target features from significant background noise. Moreover, traditional convolutions depend on static geometric kernels. This limits their directional adaptability, complicating the characterisation of uneven and indistinct lesion outlines. To overcome these limitations, we propose the SD-UNet architecture, inspired by the "coarse-to-fine" diagnostic workflow in clinical practice. Our method achieves this through two interconnected key mechanisms. First, to improve the image segmentation process, we propose a Dynamic Cognitive Diagnostic Module (DCDM). This module utilizes the initial coarse mask prediction (ROI) to dynamically adjust the weights of the small kernel convolution feature aggregation and guide subsequent edge refinement stages. This helps the model focus on relevant anatomical structures while effectively suppressing background noise. Second, to enhance the extraction of contour features, we introduce Structural Tensor Convolution

*Corresponding author.

Email addresses: 107552404008@stu.xju.edu.cn (Jiajie Wang), yerdong@xju.edu.cn (Erdong Yuan), szz@stu.xju.edu.cn (Zhaozhao Su), gaohuang@tsinghua.edu.cn (Gao Huang), 107552304108@stu.xju.edu.cn (Yuxuan Luo), yangyn5126@163.com (Yining Yang), guozitong1982@163.com (Zitong Guo), taojing_xj@163.com (Jing Tao), wljxju@xju.edu.cn (Liejun Wang)

¹These authors contributed equally to this work.

(STConv). Unlike traditional methods that rely on multilayer perceptrons (MLPs) for angle regression, STConv constructs a grouped structure tensor by calculating local spatial derivatives. By analytically solving for its principal eigenvectors, it can generate optimal orientations for dynamically rotating convolutional kernel groups. Thoroughly assessed across five public medical imaging standards, SD-UNet attains state-of-the-art performance, exhibiting solid representational stability and significant generalizability. The code is available at <https://github.com/Anonymous-Submission/SD-UNet>.

Keywords: Medical image segmentation, Dynamic cognitive diagnosis, Structure tensor convolution, Parameter-space modulation, Lesion localization

1. Introduction

Medical picture segmentation is an essential element of computer-aided diagnosis systems, significantly contributing to clinical assessment, surgical planning, and therapy evaluation. By offering objective, pixel-level delineation of diseased areas, it markedly improves diagnostic efficiency and reduces subjective human errors. Despite the significant advancements in feature learning achieved by convolutional neural networks (CNNs) [1, 2] and vision transformers (ViTs) [3, 4], these traditional architectures continue to encounter difficulties in managing adjacent anatomical structures or overlapping lesions. They encounter two principal challenges:

Challenge 1: Visual Ambiguity under Lesion Co-occurrence. In intricate clinical situations, lesions often have morphological similarities to adjacent healthy tissues, a condition termed cooccurrence [5], as seen in Fig. 1 (a). The pronounced visual similarity engenders semantic coupling, complicating the models’ ability to delineate the target region. In endoscopic imaging [6], tiny polyps frequently merge with mucosal folds of comparable texture, leading conventional models to mistakenly activate non-target areas and produce false positives. Conventional architectures, such as UNet++ [7] and standard Vision Transformers [8], rely on static, feed-forward encoding. Because they process all spatial locations with equal computational weight, they lack

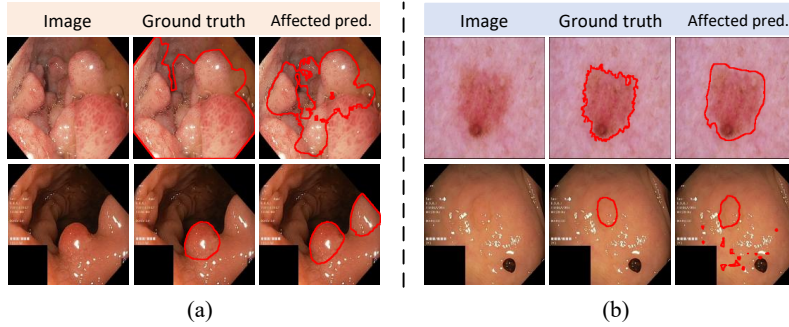


Fig. 1: Major challenges in medical image segmentation. (a) The challenge of co-occurrence. (b) The ambiguity and irregularity of boundaries.

an active mechanism to suppress background noise. Recent general purpose vision models, for instance LSNet [9] and OverLoCK [10], attempt to capture broader context via multiscale perception. However, their context-first design is primarily optimized for natural images with sharp boundaries. In medical domains characterized by low inter-class contrast, these generalist models risk discarding subtle pathological details during feature compression. Moreover, while some approaches attempt downstream filtering in the feature space, this is inherently suboptimal; the foundational convolutional weights have already been contaminated by background co-occurrence during the initial extraction phase. Clinical diagnosis typically follows a coarse-to-fine logic: first, a preliminary global screening of suspicious areas is conducted, followed by detailed local observation. Inspired by this, and drawing inspiration from the EAM [11] model’s guidance on ROIs, we embed this diagnostic workflow into our model computation framework. This mechanism effectively suppresses background noise interference before feature aggregation, thus providing a clean feature representation for subsequent edge refinement.

Challenge 2: Modeling Limitations of Fixed Geometry Sampling for Irregular Edges. Medical lesions often present with highly irregular, jagged, or bifurcated boundaries that gradually blend into normal tissue, as shown in Fig. 1 (b). Standard convolutions, limited to a fixed receptive field, often struggle to effectively capture these

multi-directional irregular contours. Existing mainstream solutions have significant limitations: multi-scale methods [12, 13] rely on predefined spatial arrangements and cannot flexibly adapt to dynamically changing edges; dynamic convolutions [14] primarily modulate features in the channel dimension, lacking modeling for spatial orientation; deformable convolutions [15] introduce unconstrained pixel-level offsets, which easily disrupt local topological continuity. Even directional convolutions such as active rotation filters [16] are constrained by rigid discrete angle quantization. To address this issue, we introduce the structure tensor [17]. In the field of directional feature analysis, the structure tensor provides a deterministic analytical solution for estimating local orientations. It can accurately extract edge orientations without introducing any additional parameters and possesses clear physical interpretability.

Based on the above analysis, we propose SD-UNet. This architecture integrates a region-aware guidance mechanism to accurately isolate lesion regions, while fusing a mathematically rigorous convolutional strategy specifically tailored to the inherent geometry of the target. The main contributions of this paper can be summarized as follows:

- We propose a Dynamic Cognitive Diagnostic Module (DCDM). This module first generates a coarse predictive mask as an initial estimate of the segmented region. Then, based on the probability map of the region of interest (ROI), it dynamically adjusts the aggregation region of the small-kernel convolution. Finally, it assists in edge refinement. This process iteratively refines and progressively improves the segmentation quality, thus ensuring the accuracy of boundary information.
- We designed Structured Tensor Convolution (STConv). It proposes a mathematically rigorous rotation-adaptive mechanism to address the limitations of fixed-direction sampling. It constructs structured tensors by computing local spatial derivatives and derives a definite physical orientation by solving for the principal eigenvectors of these tensors. Guided by this orientation, STConv can

dynamically rotate the convolution kernel, thereby achieving optimal spatial alignment with complex edges.

- **State-of-the-Art Performance:** Comprehensive assessments across five medical imaging benchmarks, specifically ISIC2017 [18], ISIC2018 [19], Kvasir [6], COVID-19 [20], and MoNuSeg [21], substantiate our methodology. SD-UNet-L sets a new benchmark, exceeding the most robust baseline by **2.87%** in average IoU and **1.70%** in average Dice.

2. Related Work

2.1. Medical Image Segmentation Models

Deep learning has facilitated significant progress in medical image segmentation [22, 23]. The original U-Net [24] established a standard for pixel-level prediction with a symmetric encoder-decoder architecture featuring skip connections. Subsequent research mostly concentrates on improving feature representation and computing performance. To enhance representational capacity, complementary designs that integrate CNNs and Transformers [1], in conjunction with multiscale subtraction networks [25], seek to eliminate background noise while emphasising prominent lesion features. To tackle the intricacies of irregular anatomical features, methods such as kMaXU [23] utilise a k-means Mask Transformer to dynamically modify spatial receptive fields. The simultaneous demand for real-time clinical implementation has generated interest in lightweight engineering. For instance, TSLDseg [26] enhances the efficacy of latent diffusion models while maintaining boundary sensitivity, while CCPNet [25] presents a pooling transformer augmented with target context to enable efficient feature aggregation.

2.2. Biomimetic Vision Models

Biological vision systems provide essential models for enhancing computational feature extraction. Initial computational models utilised the extensive field perception

of retinal peripheral vision to encompass broad context [27]. Min et al. [28] devised the Peripheral Vision Transformer to effectively segment the visual field. In a similar vein, Yang et al. [29] developed Focal Modulation Networks to reconcile intricate focal details with overarching meanings. Recently, Chen et al. [30] developed peripheral convolutions to enhance the effective receptive field in large kernel CNNs while maintaining reasonable parameter costs. Saalmann et al. [31] delineated the neural circuits of visual attention to duplicate sensory information routing, demonstrating how top-down feedback actively accentuates pertinent spatial regions. Recent architectures, such as LSNet [9], employ dual scale convolution algorithms to optimise efficiency and representational capacity, whereas OverLoCK [10] combines global overview modules with local refinement techniques. In contrast to general-purpose biomimetic models that predominantly depend on post-extraction feature attention, our methodology integrates the doctors’ coarse to fine diagnostic workflow right into the fundamental encoding phase of the network. Through parameter space modulation, we develop a specialised cognitive framework specifically designed for the identification and isolation of complicated medical lesions.

2.3. Adaptive Convolution and Geometric Modeling

Fixed grid convolutions are inadequate for modelling intricate morphologies, leading academics to investigate dynamic sampling methodologies. Dai et al. [15] proposed Deformable Convolutions for non-rigid geometric transformations, utilising learnable spatial offsets to distort the sample grid in accordance with object forms. Contemporary designs such as InternImage [32] subsequently expanded this paradigm to accomplish large-scale, data-dependent spatial modelling. Simultaneously, while pursuing wider contextual dependencies without compromising local details, RepLKNet [12] reassessed the viability of large kernel convolutions in deep networks. UniReplkNet [13] further optimised the structural reparameterization of exceptionally large kernels, explicitly enlarging the effective receptive field to improve global representations.

Another study avenue adjusts network capacity using channel-wise or weight-level ensembling. Yang et al. [33] and Chen et al. [14] introduced dynamic convolutions that linearly combine multiple parallel kernel weights based on input features, facilitating an input-dependent weight distribution. To improve representational flexibility within stringent parameter limitations, Li et al. [34] introduced Involution, which reverses the traditional design paradigm by spatially sharing weights while assigning unique kernels to various channels. Although these strategies enhance the versatility of convolution processes, they mostly concentrate on spatial scaling, coordinate translation, or channel-specific weight modification. They lack explicit modelling of orientation that preserves topology. To address this shortcoming, our approach integrates structure tensor analysis into the convolutional operator. Our STConv attains accurate orientation modelling by deriving a deterministic, closed-form analytical solution for local directions, hence preserving local geometric topology. This creates a strong correlation between the receptive field and intricate clinical parameters.

3. Methodology

As illustrated in Fig. 2, SD-UNet utilises a symmetric multistage encoder-decoder architecture. The initial three encoder levels incorporate the Dynamic Cognitive Diagnosis Module (DCDM), emphasising direct geometric modelling over general feature extraction. The DCDM functions as a region-aware guiding system, employing parameter space modulation to actively separate target lesions while efficiently suppressing concurrent background noise. This module contains a Boundary Refinement component that integrates the proposed Structure Tensor Convolution, referred to as STConv. STConv computes local spatial derivatives to derive closed-form physical orientations, rather than employing traditional feature aggregation. This approach enables adaptive edge feature extraction and allows for the accurate delineation of complex contours. The architectural flow is finalised by employing a Multi Head Self Attention bottleneck

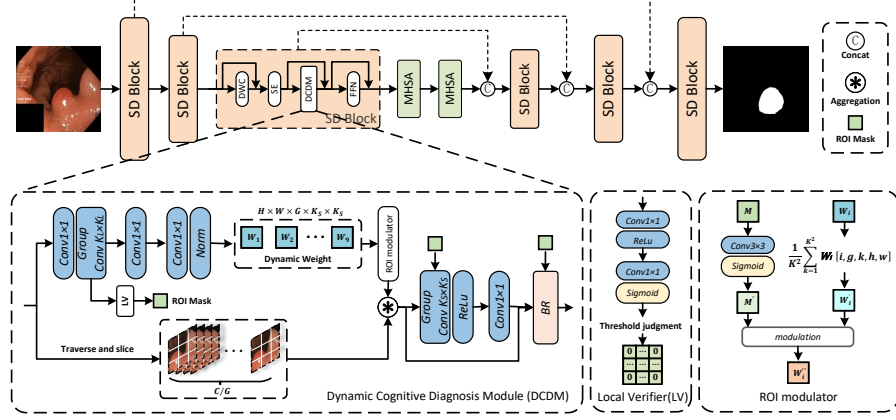


Fig. 2: Overview of the proposed SD-UNet architecture. The Local Validator (LV) explicitly anchors regions of interest by generating probabilistic masks. The Boundary Refinement (BR) module embeds Structured Tensor Convolutions (STConv) to accurately capture and reconstruct complex lesion contours. The Multi-Head Self-Attention (MHSA) module establishes long-range global context dependencies.

at the deepest level of the network to capture extensive global contextual dependencies.

3.1. Dynamic Cognitive Diagnosis Module (DCDM)

To effectively suppress interference from irrelevant background regions, we propose the Dynamic Cognitive Diagnostic Module (DCDM). This module draws upon the "coarse-to-fine" diagnostic workflow in clinical practice, overcoming the limitations of traditional feature space attention mechanisms. As shown in Figure 3, it decomposes the segmentation target into three intricate computational stages: probabilistic region localization, spatial variation feature verification, and orthogonal boundary refinement.

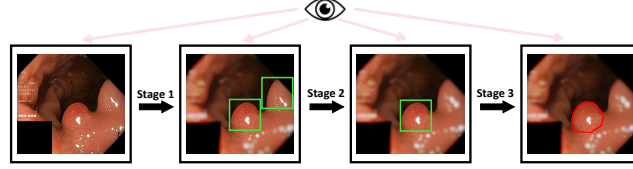


Fig. 3: The three-stage diagnostic framework: region localization, feature verification, and boundary refinement.

Region Localization. The locator module generates location-specific filter weights by integrating long-range semantic context with region-specific spatial priors. Given the

input tensor $\mathbf{X}$, a large-kernel ($K_L \times K_L$) group convolution derives the global contextual representation $\mathbf{F}_g$. Utilising a low-rank bottleneck projection, $\mathbf{F}_g$ is parameterized into the foundational dynamic filter weights $\mathcal{W} \in \mathbb{R}^{H \times W \times G \times K_S \times K_S}$.

To clearly differentiate lesion regions from healthy tissue, $\mathbf{F}_g$ is concurrently input into the Local Verifier (LV). This lightweight branch has two 1×1 convolutional layers interspersed with a ReLU activation to forecast a spatial probability map $\mathbf{P}$. The map is then binarized with a threshold a (established at 0.5) to generate a definitive ROI mask $\mathcal{M}$. Ultimately, within the ROI Modulator, the background prior $\mathcal{W}'$ is derived by executing an averaging operation on $\mathcal{W}$ throughout its spatial dimensions. Simultaneously, the binary mask $\mathcal{M}$ undergoes a 3×3 convolution followed by a Sigmoid activation. This effectively smooths the discrete boundaries, producing a differentiable soft spatial gate $\mathcal{M}'$. The ultimate dynamic weights $\mathcal{W}''$ are generated using an exact convex combination:

$$\mathcal{W}'' = \mathcal{W} \odot \mathcal{M}' + \mathcal{W}' \odot (1 - \mathcal{M}') \quad (1)$$

where $\odot$ denotes element-wise multiplication.

Feature Verification. Following the parameter modulation, the Feature Inspector functions as a semantic verification mechanism. Unlike standard convolutions that share invariant kernels globally, this stage executes a spatially-variant dynamic convolution, explicitly conditioned on the localized weights $\mathcal{W}''$. The location-specific tensor aggregation is mathematically defined as:

$$\mathbf{F}_{agg}(p) = \sum_{q \in \Omega(p)} \mathcal{W}''_p(p - q) \odot \mathbf{X}(q) \quad (2)$$

where p and q denote spatial coordinates, and $\Omega(p)$ represents the local receptive field. This operation structurally aligns the feature manifold with the dynamic geometric context. To further guarantee representation purity, the aggregated features are strictly

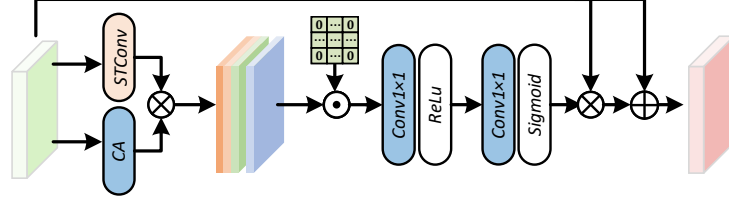


Fig. 4: Detailed design of the Boundary Refinement (BR) module.

regularized by the shared binary mask $\mathcal{M}$ and corrected via a local verifier network Φ_{ver} :

$$\mathbf{X}' = \mathbf{F}_{agg} + \Phi_{ver}(\mathbf{F}_{agg} \odot \mathcal{M}) \quad (3)$$

Boundary Refinement. To complement the semantic localization, the Boundary Refinement (BR) module focuses on the topological precision of lesion contours. We introduce an orthogonal feature calibration strategy. Specifically, the verified feature $\mathbf{X}'$ is simultaneously processed by two parallel branches. The Structure Tensor Convolution (STConv) extracts physically grounded boundary orientations, while a Channel Attention (CA) module recalibrates semantic inter-dependencies. The resulting synergistic geometric feature representation is formulated via element-wise multiplication as:

$$\mathbf{F}_{cal} = \text{STConv}(\mathbf{X}') \odot \text{CA}(\mathbf{X}') \quad (4)$$

To systematically suppress high-frequency background artifacts, we cascade the single-source mask guidance. $\mathbf{F}_{cal}$ is spatially gated by the resolution-aligned ROI mask $\mathcal{M}$ and subsequently projected through an edge-aware enhancement network Φ_{edge} . As shown in Fig. 4, Φ_{edge} employs a bottleneck architecture comprising two 1×1 convolutions to aggregate cross-channel information. This process is mathematically defined as:

$$\mathbf{A}_{edge} = \sigma(\mathbf{W}_2 * \delta(\mathbf{W}_1 * (\mathbf{F}_{cal} \odot \mathcal{M}))) \quad (5)$$

where $*$ denotes the convolution operation, and δ and σ represent the ReLU and

Sigmoid activation functions, respectively. To realize this bottleneck, the convolutional weights are parameterized as $\mathbf{W}_1 \in \mathbb{R}^{\frac{C}{r} \times C \times 1 \times 1}$ and $\mathbf{W}_2 \in \mathbb{R}^{1 \times \frac{C}{r} \times 1 \times 1}$, where r denotes the channel reduction ratio. This sequential projection yields a precise single-channel spatial activation map $\mathbf{A}_{edge}$. Finally, the original features are selectively enhanced to restore intricate boundary topologies via a residual spatial attention mechanism:

$$\mathbf{X}_{out} = \mathbf{X}' + \mathbf{X}' \odot \mathbf{A}_{edge} \quad (6)$$

Remark. Traditional attention strategies directly adjust the feature space. This method frequently leads to inadequate boundary delineation because to intrinsic feature entanglement. The proposed DCDM functions within a unique cognitive diagnosing framework. The soft mask $\mathcal{M}$ serves as a continuous diagnostic prior produced by the probabilistic region localisation phase. This prior is systematically disseminated across three orthogonal computational dimensions, rather than merely functioning as a background suppressor. Initially, in parameter-space routing as described in Eq. 1, $\mathcal{M}$ generates dynamic filters tailored to specific locations. Secondly, in the feature-space verification outlined in Eq. 3, $\mathcal{M}$ functions as a spatial regularizer for the attenuation of noise. Third, in the topology-space refinement in Eq. 5, $\mathcal{M}$ restricts the border enhancement to concentrate just on transitions between lesions and backgrounds. This mechanism of progressively refining the coarse segmentation ensures that feature extraction is an interpretable and rigorously mathematically constrained optimization process. DCDM is not simply a stacking of modules, but rather a clear theoretical bridge between global spatial localization and local geometric reconstruction.

3.2. Structure Tensor Convolution

To transcend the static sampling paradigm of traditional convolutions and explicitly model the multi-oriented, irregular edges in medical images, we propose the Structure Tensor Convolution (STConv) module. As shown in Fig. 5, unlike existing dynamic

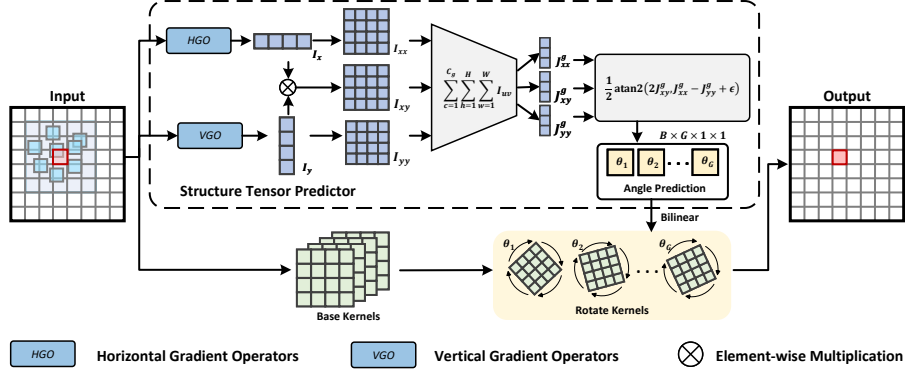


Fig. 5: Detailed design of the STConv.

convolutions that rely on heuristic MLP-based angle regression, STConv operates as a mathematically grounded, zero-parameter geometric computation. It computes local spatial derivatives to analytically deduce the macro-dominant physical orientation for dynamic kernel steering.

Given an input feature map $\mathbf{X} \in \mathbb{R}^{B \times C \times H \times W}$, we first partition the channels into $G = \min(\text{groups}, C)$ mutually exclusive groups, with each group containing $C_g = C/G$ channels. To completely eliminate the dependency on opaque neural layers, we formulate a zero-parameter Structure Tensor Predictor. As illustrated in Fig. 6, we perform directional gradient extraction via a Horizontal Gradient Operator (HGO) and a Vertical Gradient Operator (VGO). Using fixed orthogonal derivative priors $\mathbf{S}_x$ and $\mathbf{S}_y$, we compute the first-order spatial derivatives of the input features:

$$\mathbf{I}_x = \mathbf{X} * \mathbf{S}_x, \quad \mathbf{I}_y = \mathbf{X} * \mathbf{S}_y \quad (7)$$

where $*$ denotes the filtering operation. To encapsulate the inherent geometric texture, we formulate the local structure tensor components by calculating the gradient energies and their cross-correlations: $I_{xx} = \mathbf{I}_x \odot \mathbf{I}_x$, $I_{yy} = \mathbf{I}_y \odot \mathbf{I}_y$, and $I_{xy} = \mathbf{I}_x \odot \mathbf{I}_y$. The orthogonal derivative operators $\mathbf{S}_x$ and $\mathbf{S}_y$ are definitively constant. If let to update freely during backpropagation, they would deteriorate into unregulated feature extractors. As a result,

the components (I_{xx}, I_{yy}, I_{xy}) would forfeit their specific physical significance as a local gradient covariance matrix.

To guarantee that the orientation is resilient to high-frequency speckle noise characteristic of medical imaging, we globally aggregate these tensor components inside each discrete group $g \in \{1, \dots, G\}$. The group-wise tensor elements are derived by aggregating across the spatial dimensions $(H \times W)$ and the intra-group channels (C_g) :

$$J_{uv}^g = \sum_{c=1}^{C_g} \sum_{h=1}^H \sum_{w=1}^W J_{uv}^{(c,h,w)}, \quad u, v \in \{x, y\} \quad (8)$$

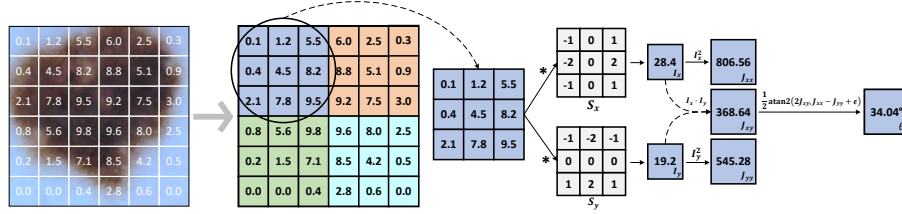


Fig. 6: Computational workflow of the proposed STConv. The procedure commences with the extraction of a 3×3 local patch from the boundary region to compute spatial gradients utilising HGO and VGO. Subsequent to this phase, the resultant first-order derivatives are transformed into second-order structure tensor energy components.

Based on these aggregated energies, the structure tensor for the g -th group is defined. By analytically solving for the principal eigenvector of this tensor, we derive the closed-form physical orientation θ_g that maximizes the local gradient variance:

$$\theta_g = \frac{1}{2} \text{atan2} \left(2J_{xy}^g, J_{xx}^g - J_{yy}^g + \epsilon \right) \quad (9)$$

where $\epsilon = 10^{-8}$ serves as a numerical stabiliser. This analytical approach inherently derives an absolute, optimal rotation angle $\theta_g \in [-\pi/2, \pi/2]$ for each group. It depends solely on physical structural priors, entirely circumventing the necessity for learnable parameters. Figure 6 explicitly illustrates this geometric derivation utilising a representative 3×3 local border patch. Standard convolutions demonstrate a pronounced inductive bias favouring the Cartesian grid. Although this enables them to efficiently

extract axis-aligned features, they are intrinsically prone to grid aliasing when navigating oblique edges. By rotating the kernel to align with the predominant macro orientation, STConv mathematically converts arbitrarily oriented anatomical edges into axis-aligned features about the kernel’s local coordinate system.

For each sample b and group g , we dynamically align the base kernel slice $\mathcal{K}_{\text{base}}^{(g)} \in \mathbb{R}^{C_g \times C_g \times k \times k}$ to the calculated dominant orientation $\theta_{b,g}$. We establish a spatial transformation that translates the target grid coordinates (x, y) in the rotated kernel space to the source coordinates (x', y') in the base kernel space:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta_{b,g} & -\sin \theta_{b,g} \\ \sin \theta_{b,g} & \cos \theta_{b,g} \end{bmatrix} \begin{bmatrix} x - c_x \\ y - c_y \end{bmatrix} + \begin{bmatrix} c_x \\ c_y \end{bmatrix} \quad (10)$$

where $(c_x, c_y) = (\frac{k-1}{2}, \frac{k-1}{2})$ represents the kernel centre. The rotated kernel values $\mathcal{K}_{\text{rot}}^{(b,g)}$ are calculated using differentiable bilinear interpolation derived from the inverse mapping. This guarantees that the kernel remains spatially continuous and differentiable during the optimisation process. The geometrically aligned kernels $\mathcal{K}_{\text{rot}} \in \mathbb{R}^{B \times C \times C_g \times k \times k}$ are then utilised by grouped convolution. By methodically directing the receptive field to correspond with the prevailing local orientation, STConv adeptly maintains fine border details while alleviating the gradient dilution commonly experienced at complicated multi-directional edges.

Remark. Most current dynamic convolutions rely on parameterized multilayer perceptrons to model geometric transformations. Medical photos can exhibit significant speckle noise and poor contrast artefacts. In demanding situations, these unconstrained networks readily overfit to misleading local gradients rather than authentic structural characteristics. The suggested STConv utilises a deterministic physical prior to mitigate this problem. The combined tensor components, namely J_{xx}^g , J_{yy}^g , and J_{xy}^g , constitute a local gradient covariance matrix that encapsulates the inherent dimensionality of the feature manifold.

3.3. Loss Function

In the training phase, we leverage pixel-level supervision on $\hat{y}_i$ to achieve rapid convergence. Specifically, the overall loss function is formulated as:

$$\mathcal{L} = \underbrace{\lambda_1 \left(-\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \right)}_{\mathcal{L}_{BCE}} + \underbrace{\lambda_2 \left(1 - \frac{2|X \cap Y|}{|X| + |Y|} \right)}_{\mathcal{L}_{Dice}} \quad (11)$$

where N is the total number of samples, y_i is the ground truth label, and $\hat{y}_i$ is the predicted probability. $|X|$ and $|Y|$ denote the ground truth and prediction sets, respectively. $\mathcal{L}_{BCE}$ and $\mathcal{L}_{Dice}$ represent the binary cross-entropy [35] and Dice loss [36], respectively. λ_1 and λ_2 are hyper-parameters used to balance the corresponding loss terms.

4. Experiments

4.1. Datasets

To thoroughly assess the segmentation efficacy of the proposed SD-UNet architecture, we performed extensive tests on five separate public medical picture benchmarks. The meticulously curated datasets comprise ISIC2017 [18], ISIC2018 [19], Kvasir [6], COVID-19 [20], and MoNuSeg [21]. To ensure stringent consistency during the evaluation process, each dataset was divided using the same ratio, specifically designating 80% of the samples for training and the remaining 20% for testing. The quantitative assessment utilises five established indicators to provide a comprehensive evaluation of boundary demarcation and regional overlap. These metrics include Intersection over Union (IoU), the Dice similarity coefficient, Recall, Precision, and the 95th percentile Hausdorff Distance (HD95).

4.2. Implementation Details

The proposed network and all baseline models are implemented using the PyTorch framework, and all experiments were performed on a single NVIDIA RTX A6000

GPU. To ensure absolute fairness in the comparison, all models used uniform training hyperparameters. Input images were uniformly adjusted to 256×256 resolution. The optimization process used the AdamW optimizer with weight decay set to 1×10^{-2} and ϵ set to 1×10^{-8} . The batch size was set to 8, and training lasted for 200 epochs. The initial learning rate was set to 1×10^{-3} , and a cosine annealing strategy with 50 iterations was used for dynamic decay to avoid getting trapped in local optima. In addition, conventional data augmentation techniques such as horizontal flipping, vertical flipping, and random rotation were used during training to improve the spatial robustness of the model.

Table 1: The quantitative comparison among different medical segmentation networks evaluated on the ISIC2017, ISIC2018, and Kvasir datasets. The best and second-best results are highlighted in **bold** and underlined, respectively.

Network	Params↓	GFLOPs↓	ISIC2017		ISIC2018		Kvasir	
			IoU (%)↑	Dice (%)↑	IoU (%)↑	Dice (%)↑	IoU (%)↑	Dice (%)↑
heavy-weight medical network								
U-Net (MICCAI'15) [24]	34.52 M	65.52	72.48	84.05	80.10	88.91	73.30	84.59
TransUNet (ArXiv'21) [37]	105.32 M	38.52	73.73	84.88	88.44	93.86	80.98	89.49
UCTransNet (AAAI'22) [38]	66.24 M	32.98	72.79	84.25	81.90	90.02	90.15	94.82
CMU-Net (ISBI'23) [39]	49.93 M	91.25	71.56	83.42	80.53	91.97	89.90	94.68
VM-UNet (ACM TOMM'24) [40]	44.27 M	7.56	67.75	80.78	82.07	90.15	88.71	91.08
EMCAD (CVPR'24) [41]	26.76 M	5.60	64.46	78.39	80.71	89.32	71.73	83.53
RWKV-UNet (ArXiv'25) [42]	18.42 M	13.28	75.16	85.82	88.01	93.54	89.11	94.24
LGFVM-UNet (EAAI'26) [43]	58.83 M	11.89	70.89	82.97	81.13	89.58	75.32	85.92
light-weight medical network								
EGE-Unet (MICCAI'23) [44]	0.072 M	0.045	70.73	82.85	84.82	91.79	72.44	84.02
CMUNeXt (ISBI'24) [45]	3.14 M	7.41	73.33	84.61	87.81	92.99	88.85	94.09
TinyU-Net (MICCAI'24) [46]	0.48 M	1.66	73.66	84.83	88.40	92.71	73.04	84.42
MobileUViT (ACM MM'25) [47]	6.21 M	10.43	76.14	85.59	83.69	91.12	87.95	93.59
MAMBAU-lite (ICISN'25) [48]	0.39 M	0.93	71.74	86.59	86.25	92.61	79.66	88.67
H-VMUNet (Neurocomput'25) [49]	8.97 M	0.76	75.47	86.17	85.51	92.18	85.53	89.82
MK-UNet (ICCV'25) [49]	0.32 M	0.33	76.03	86.38	89.01	94.19	88.18	93.72
SD-UNet-B (Ours)	4.71 M	12.88	76.61	86.76	88.76	93.43	91.09	95.34
SD-UNet-L (Ours)	5.83 M	17.94	77.98	87.63	89.61	93.83	91.72	95.68

4.3. Comparison With State of the Arts

This section provides a detailed comparison between SD-UNet and established state of the art methods, specifically U-Net [24], TransUNet [37], UCTransNet [38], CMU-Net [39], VM-UNet [40], EMCAD [41], RWKV-UNet [42], EGE-UNet [44], CMUNeXt [45], TinyU-Net [46], MobileUViT [47], MAMBAU-lite [48], H-VMUNet [49], MK-UNet [49], and LGFVM-UNet [43]. To facilitate a structured assessment,

Table 2: The quantitative comparison among different medical segmentation networks evaluated on the COVID-19 and MoNuSeg datasets.

Network	Params↓	GFLOPs↓	COVID-19		MoNuSeg	
			IoU (%)↑	Dice (%)↑	IoU (%)↑	Dice (%)↑
<i>heavy-weight medical network</i>						
U-Net (MICCAI'15) [24]	34.52 M	65.52	67.82	80.79	36.05	53.00
TransUNet (ArXiv'21) [37]	105.32 M	38.52	77.41	86.29	67.42	80.54
UCTransNet (AAAI'22) [38]	66.24 M	32.98	75.49	85.58	39.71	56.84
CMU-Net (ISBI'23) [39]	49.93 M	91.25	54.03	70.16	70.34	82.59
VM-UNet (ACM TOMM'24) [40]	44.27 M	7.56	69.99	82.51	66.89	80.16
EMCAD (CVPR'24) [41]	26.76 M	5.60	68.88	82.67	41.20	58.35
RWKV-UNet (ArXiv'25) [42]	18.42 M	13.28	72.95	84.36	62.71	77.08
LGFVM-UNet (EAAI'26) [43]	58.83 M	11.89	66.47	79.95	59.20	74.37
<i>light-weight medical network</i>						
EGE-UNet (MICCAI'23) [44]	0.072 M	0.045	64.25	78.24	54.25	70.34
CMUNeXt (ISBI'24) [45]	3.14 M	7.41	55.16	71.10	70.09	82.41
TinyU-Net (MICCAI'24) [46]	0.48 M	1.66	55.94	71.75	66.03	79.54
MobileUViT (ACM MM'25) [47]	6.21 M	10.43	52.07	68.48	67.98	80.94
MAMBAU-lite (ICISN'25) [48]	0.39 M	0.93	73.66	81.64	49.82	66.50
H-VMUNet (Neurocomput'25) [49]	8.97 M	0.76	66.57	80.09	55.06	71.02
MK-UNet (ICCV'25) [49]	0.32 M	0.33	80.66	89.30	66.37	79.78
SD-UNet-B(Ours)	4.71 M	12.88	79.33	88.84	72.91	84.34
SD-UNet-L(Ours)	5.83 M	17.94	81.84	90.01	73.43	84.68

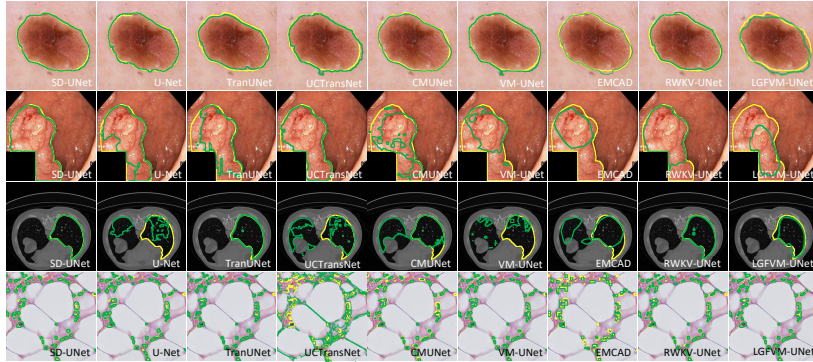


Fig. 7: Qualitative visual comparison between SD-UNet and other heavy-weight networks across four distinct imaging modalities.

we categorize these baselines into heavyweight and lightweight architectures.

Quantitative Results. Tables 1 and 2 present the assessment metrics for our methodology. We evaluate two variants, SD-UNet-B and SD-UNet-L, against the previously identified fifteen baselines. SD-UNet-L attains an average IoU of 82.92% and an average Dice score of 90.37%, establishing a new benchmark among the

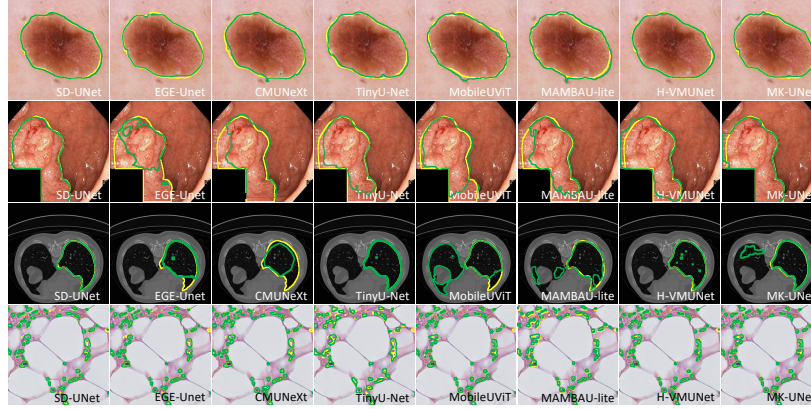


Fig. 8: Qualitative visual comparison between SD-UNet and other light-weight networks across four distinct imaging modalities.

evaluated datasets. In the Kvasir dataset, which features lesions with ambiguous boundaries and significant background noise, our model demonstrates a substantial improvement, achieving 91.72% in IoU and 95.68% in Dice. In the ISIC2017 and ISIC2018 datasets, SD-UNet-L attains superior IoU scores of 77.98% and 89.61%, respectively, while also demonstrating very competitive Dice. In the COVID-19 dataset, SD-UNet-L exceeds the primary lightweight competitor, MK-UNet, by 1.18% in IoU. Similarly, utilising the MoNuSeg dataset, our model surpasses the robust CMU-Net technique by 3.09% in IoU and 2.09% in Dice.

Qualitative Results. As shown in Fig. 7 and Fig. 8, SD-UNet exhibits a superior capacity to mitigate background artefacts and restore intricate topological details in comparison to both heavyweight and lightweight networks. As a result, it generates prediction masks that closely correspond to the ground truth, even in intricate areas.

Average Performance and Efficiency. Table 3 displays the average IoU and Dice scores for all five datasets, together with indicators of model complexity, emphasising the trade-off between performance and computational cost. The fundamental model, SD-UNet-B, utilising merely 4.71 M parameters, achieves an average IoU of 81.74%, surpassing the performance of the foremost lightweight competitor, MK-UNet. The

Table 3: Quantitative comparison of average segmentation performance across all evaluated datasets. The baselines are categorized into heavy-weight and light-weight architectures.

Network	Params↓	GFLOPs↓	Average	
			IoU (%)↑	Dice (%)↑
<i>heavy-weight medical network</i>				
U-Net (MICCAI'15) [24]	34.52 M	65.52	65.95	78.27
TransUNet (ArXiv'21) [37]	105.32 M	38.52	77.60	87.01
UCTransNet (AAAI'22) [38]	66.24 M	32.98	72.01	82.30
CMU-Net (ISBI'23) [39]	49.93 M	91.25	73.27	84.56
VM-UNet (ACM TOMM'24) [40]	44.27 M	7.56	75.08	84.94
EMCAD (CVPR'24) [41]	26.76 M	5.60	65.40	78.45
RWKV-UNet (ArXiv'25) [42]	18.42 M	13.28	77.59	87.01
LGFVM-UNet (EAAI'26) [43]	58.83 M	11.89	70.60	82.56
<i>light-weight medical network</i>				
EGE-Unet (MICCAI'23) [44]	0.072 M	0.045	69.30	81.45
CMUNeXt (ISBI'24) [45]	3.14 M	7.41	75.05	85.04
TinyU-Net (MICCAI'24) [46]	0.48 M	1.66	71.41	82.65
MobileUViT (ACM MM'25) [47]	6.21 M	10.43	73.57	83.94
MAMBAU-lite (ICISN'25) [48]	0.39 M	0.93	72.23	83.20
H-VMUNet (Neurocomput'25) [49]	8.97 M	0.76	73.63	83.86
MK-UNet (ICCV'25) [49]	0.32 M	0.33	80.05	88.67
SD-UNet-B(Ours)	4.71 M	12.88	81.74	89.74
SD-UNet-L(Ours)	5.83 M	17.94	82.92	90.37

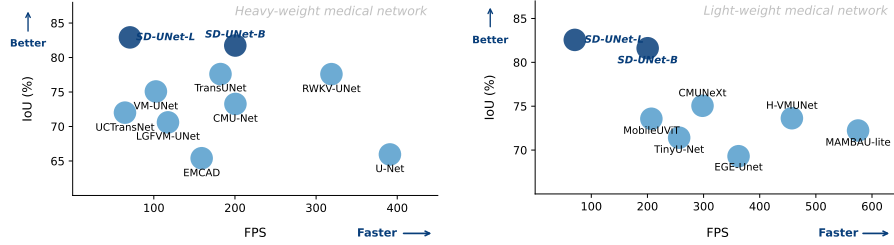


Fig. 9: Performance v.s. FPS across heavy- and light-weight medical. We report the average IoU of four datasets.

larger variant, SD-UNet-L, attains the highest position with an average IoU of 82.92% and a Dice score of 90.37%. It far surpasses heavyweight designs. Additionally, Fig. 9 maps the performance against inference speed, measured in frames per second or FPS. The scatter plot shows that SD-UNet achieves leading average segmentation accuracy while still fully meeting the requirements of real-time clinical deployment in terms of FPS.

Zero-Shot Cross-Dataset Generalization. To rigorously assess resilience and

Table 4: All models are trained on either the Kvasir or ISIC2017 source dataset and directly tested on the unseen CVC-ClinicDB or PH2 target dataset.

Network	Params↓	GFLOPs↓	Kvasir → CVC-ClinicDB		ISIC2017 → PH2	
			IoU (%)↑	Dice (%)↑	IoU (%)↑	Dice (%)↑
<i>heavy-weight medical network</i>						
U-Net (MICCAI'15) [24]	34.52 M	65.52	56.47	70.83	81.86	90.02
TransUNet (ArXiv'21) [37]	105.32 M	38.52	59.88	74.90	84.89	91.83
UCTransNet (AAAI'22) [38]	66.24 M	32.98	64.95	<u>78.75</u>	83.94	91.27
CMU-Net (ISBI'23) [39]	49.93 M	91.25	53.81	69.97	82.24	90.25
VM-UNet (ACM TOMM'24) [40]	44.27 M	7.56	54.17	66.46	79.10	88.33
EMCAD (CVPR'24) [41]	26.76 M	5.60	57.53	74.43	72.76	84.23
RWKV-UNet (ArXiv'25) [42]	18.42 M	13.28	44.38	61.47	84.69	91.71
LGFVM-UNet (EAAI'26) [43]	58.83 M	11.89	43.93	61.05	80.97	89.48
<i>light-weight medical network</i>						
EGE-Unet (MICCAI'23) [44]	0.072 M	0.045	44.37	61.47	83.74	91.15
CMUNeXt (ISBI'24) [45]	3.14 M	7.41	61.38	76.07	83.62	91.08
TinyU-Net (MICCAI'24) [46]	0.48 M	1.66	53.60	69.79	83.38	90.94
MobileUViT (ACM MM'25) [47]	6.21 M	10.43	58.68	73.96	75.06	85.76
MAMBAU-lite (ICISN'25) [48]	0.39 M	0.93	37.71	54.77	80.96	89.48
H-VMUNet (Neurocomput'25) [49]	8.97 M	0.76	55.44	71.33	83.59	91.06
MK-UNet (ICCV'25) [49]	0.32 M	0.33	57.68	73.16	84.88	91.82
SD-UNet-B(Ours)	4.71 M	12.88	<u>65.07</u>	78.69	86.89	92.98
SD-UNet-L(Ours)	5.83 M	17.94	67.61	78.91	<u>86.41</u>	<u>92.71</u>

out of distribution generalization, we conducted zero shot cross dataset evaluations, detailed in Tab. 4. Under this protocol, models trained exclusively on a source domain are directly evaluated on an unseen target domain without any fine tuning. In the first scenario transferring from Kvasir to CVC ClinicDB, despite significant variations in endoscopic lighting and polyp morphologies, SD-UNet-L maintains diagnostic stability. It achieves an IoU of 67.61% and a Dice score of 78.91%. This result strongly surpasses heavily parameterized baselines such as UCTransNet, which experiences a sharp performance drop to an IoU of 64.95. In the skin lesion transfer task from ISIC2017 to PH2, the base variant SD-UNet-B secures the highest IoU of 86.89% and a Dice score of 92.98%, outperforming all other competitors. These cross domain results confirm the core design philosophy of our architecture.

4.4. Ablation Studies

To demonstrate the effectiveness of our proposed SD-UNet, we conducted comprehensive ablation experiments on all suggested modules within the DCDM, which will

Table 5: Ablation study on DCDM on the **Kvasir** dataset.

Modules	Params↓	GFLOPs↓	Kvasir			
			IoU(%)↑	Dice(%)↑	Rec.(%)↑	Prec.(%)↑
Baseline	5.43 M	17.14	87.91	93.57	92.53	94.63
w/o ROI	5.82 M	17.93	91.00	95.29	94.28	96.31
w/o Edge	5.82 M	17.93	91.37	95.49	95.31	95.67
Add DCDM	5.83 M	17.94	92.48	96.09	95.25	96.25

Table 6: Ablation study on DCDM on the **COVID-19** dataset.

Modules	Params↓	GFLOPs↓	COVID-19			
			IoU(%)↑	Dice(%)↑	Rec.(%)↑	Prec.(%)↑
Baseline	5.43 M	17.14	75.79	86.23	87.93	84.60
w/o ROI	5.82 M	17.93	75.64	86.13	86.23	86.04
w/o Edge	5.82 M	17.93	78.88	88.19	88.99	87.41
Add DCDM	5.83 M	17.94	81.84	90.01	89.94	90.08

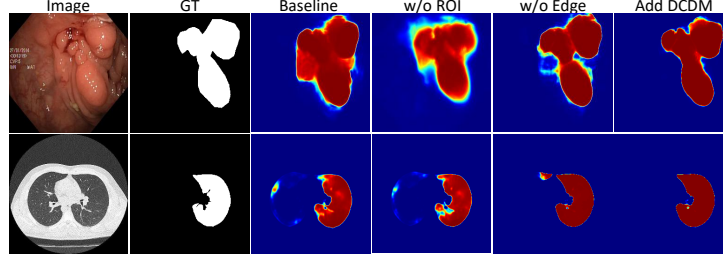


Fig. 10: Visual comparison of feature heatmaps of DCDM under different ablation configurations.

be detailed in the following sections.

4.4.1. Ablation Study on DCDM

To evaluate the contribution of the Dynamic Cognitive Diagnosis Module, we conducted ablation experiments using the Kvasir and COVID-19 datasets. The baseline architecture was compared against variants that progressively removed the ROI modulation and the edge aware components. As detailed in Tables 5 and 6, the full module achieves the best overall performance, yielding IoU scores of 92.48% and 81.84% on the respective datasets. Fig. 10 illustrates this qualitative advantage. As shown, the full configuration generates a well-defined and highly focused attention map, effectively suppressing background noise and edge blurring issues common in the baseline and some variants.

4.4.2. Ablation Study of STConv

To thoroughly evaluate the performance enhancements realised by the Structure Tensor Convolution (STConv), we categorise its effects into three separate categories. This encompasses a comparative analysis of conventional convolution operations, an assessment of directional modelling mechanisms, and an optimisation of orientation group configurations.

Table 7: Ablation study of the proposed STConv module across four medical image datasets. The baseline methods are logically categorized into Spatial & Deformable mechanisms and Dynamic & Feature-based mechanisms.

Methods	Params ↓	GFLOPs↓	ISIC2018		Kvasir		COVID-19		MoNuSeg	
			IoU(%)↑	HD95↓	IoU(%)↑	HD95↓	IoU(%)↑	HD95↓	IoU(%)↑	HD95↓
<i>Spatial & Deformable Convolutions</i>										
Conv[50]	5.43 M	17.14	87.72	3.97	89.90	2.54	76.31	12.73	68.29	2.41
DeformableConv[15]	5.67 M	19.70	84.34	4.17	83.01	6.64	68.85	19.57	64.92	2.88
GConv[51]	5.65 M	18.83	88.00	2.53	88.70	4.50	78.78	10.88	70.08	<u>1.95</u>
LSKConv[52]	5.87 M	20.26	84.24	4.53	<u>91.43</u>	3.43	<u>79.78</u>	14.61	71.74	1.99
<i>Dynamic & Feature-based Convolutions</i>										
CondConv[33]	5.47 M	17.95	86.07	3.63	90.33	4.69	71.83	17.56	68.98	2.25
DynamicConv[14]	5.47 M	17.95	87.97	4.97	89.57	2.75	76.17	12.72	70.17	1.99
ORConv[16]	5.47 M	17.93	87.96	4.81	90.03	5.17	75.83	13.27	67.04	2.24
DSConv[53]	5.79 M	18.53	88.58	2.17	90.59	4.55	77.12	12.45	71.24	2.07
DRConv[54]	8.48 M	32.86	<u>89.31</u>	1.97	90.30	4.28	78.71	12.69	71.51	2.05
SD-UNet-B(Ours)	4.71 M	12.88	88.76	2.69	91.09	<u>1.74</u>	79.33	12.32	<u>72.91</u>	2.12
SD-UNet-L(Ours)	5.83 M	17.94	89.61	<u>2.07</u>	91.72	1.66	81.84	<u>11.38</u>	73.43	1.93

Comparison with Mainstream Convolutions. Tab. 7 and Fig. 11 presents an extensive evaluation across four diverse medical datasets. To isolate the source of improvement, we benchmark STConv against two major categories of advanced convolutional operators. Initially, STConv demonstrates a distinct superiority in topological stability when contrasted with spatial and deformable convolutions. Unconstrained spatial transformation techniques, such as DeformableConv, experience significant performance decline on the Kvasir dataset, with their IoU decreasing to 83.01% and the HD95 error escalating to 6.64. In the COVID-19 dataset, STConv exceeds the conventional convolutional baseline by 5.53% in IoU and outperforms the comparable big kernel LSKConv by 2.06%. Secondly, when compared to dynamic and feature-based convolutions, STConv exhibits markedly superior parameter efficiency. Techniques such as DRConv attain a competitive IoU of 89.31% on ISIC2018. Nonetheless, their

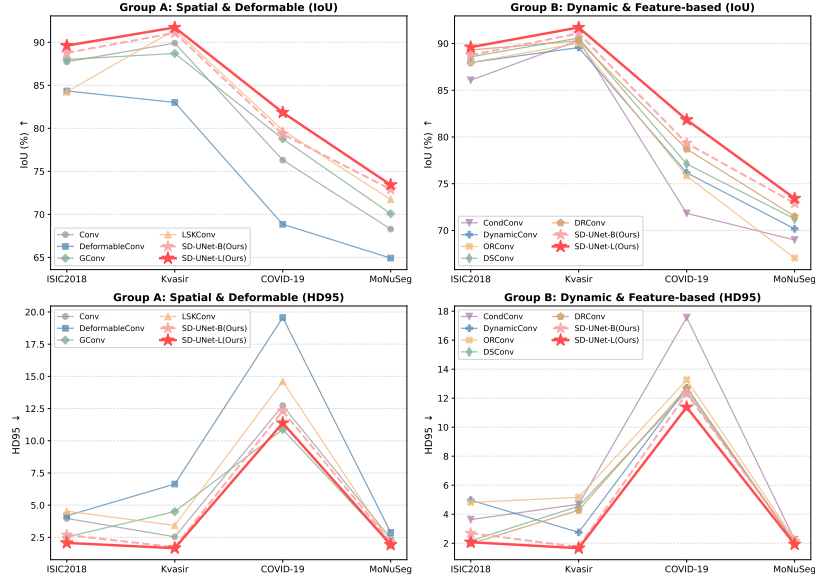


Fig. 11: Comprehensive quantitative comparison of the proposed SD-UNet variants against state-of-the-art methods across four benchmark datasets.

dependence on channel-wise weight ensembling results in a significant computational expense of 32.86 GFLOPs and 8.48 million parameters. SD-UNet-L surpasses this method while necessitating almost half the processing resources, particularly 17.94 GFLOPs. In the Kvasir dataset, standard grouping convolutions and dynamic convolutions exhibit difficulties with indistinct mucosal folds, resulting in elevated HD95 scores of 4.50 and 2.75, respectively. SD-UNet-L significantly diminishes this inaccuracy to 1.66.

Table 8: Ablation study of directional modeling mechanisms on the **Kvasir** dataset.

Model Variants	Params ↓	GFLOPs ↓	IoU (%) ↑	Dice (%) ↑	HD95 ↓
MLP-STConv ($\pm 45^\circ$)	5.91 M	17.95	91.53	95.58	2.65
MLP-STConv ($\pm 90^\circ$)	5.91 M	17.95	90.92	95.24	1.96
STConv ($\pm 45^\circ$)	5.83 M	17.94	90.96	95.26	3.49
STConv ($\pm 90^\circ$)	5.83 M	17.94	91.72	95.68	1.66

Effectiveness of Directional Modeling. We conducted a comprehensive analysis of the internal structure of STConv by evaluating the feature mapping technique and the

Table 9: Ablation study of directional modeling mechanisms on the COVID-19 dataset.

Model Variants	Params↓	GFLOPs↓	IoU (%)↑	Dice (%)↑	HD95↓
MLP-STConv ($\pm 45^\circ$)	5.91 M	17.95	79.00	88.27	13.11
MLP-STConv ($\pm 90^\circ$)	5.91 M	17.95	79.51	88.59	9.41
STConv ($\pm 45^\circ$)	5.83 M	17.94	75.97	86.35	12.86
STConv ($\pm 90^\circ$)	5.83 M	17.94	81.84	90.01	11.38

angular range of orientation modeling. As shown in Tables 8 and 9, STConv reduces the number of parameters from 5.91 M to 5.83 M while continuously improving boundary preservation. On the COVID-19 dataset, the $\pm 90^\circ$ STConv configuration achieves an IoU of 81.84%, significantly outperforming the 79.51% of the MLP variant. Furthermore, we extended the orientation modeling range from $\pm 45^\circ$ to $\pm 90^\circ$. This angular expansion significantly reduces boundary delineation error, lowering the HD95 score from 3.49 to 1.66 on the Kvasir dataset.

Table 10: Ablation study on the number of orientation groups (G) in STConv on the Kvasir dataset.

Groups (G)	Params↓	GFLOPs↓	IoU (%)↑	Dice (%)↑	HD95 ↓
$G = 1$	8.31 M	17.94	90.75	95.15	3.56
$G = 2$	6.89 M	17.94	91.46	95.54	3.13
$G = 4$	6.18 M	17.94	91.66	95.65	2.37
$G = 8$	5.83 M	17.94	91.72	95.68	1.66
$G = 16$	5.65 M	17.94	87.21	93.15	2.89
$G = C$	5.48 M	17.94	87.89	93.55	5.31

Table 11: Ablation study on the number of orientation groups (G) in STConv on the COVID-19 dataset.

Groups (G)	Params ↓	GFLOPs ↓	IoU (%)↑	Dice (%)↑	HD95 ↓
$G = 1$	8.31 M	17.94	79.51	88.58	13.05
$G = 2$	6.89 M	17.94	77.93	87.59	13.40
$G = 4$	6.18 M	17.94	77.59	87.38	12.94
$G = 8$	5.83 M	17.94	81.84	90.01	11.38
$G = 16$	5.65 M	17.94	78.08	87.69	11.87
$G = C$	5.48 M	17.94	73.32	84.60	15.20

Effectiveness of Orientation Groups. Tables 10 and 11 show that reducing G to 1 or 2 significantly increases the number of parameters to 8.31 M and limits the ability to represent directions, resulting in insufficient boundary localization. In contrast,

substantially increasing G to 16, or applying complete channel-wise grouping where G equals C , leads to a notable decline in performance. In the Kvasir dataset, increasing G to 16 decreases the IoU from 91.72% to 87.21% and raises the HD95 error to 2.89.

5. Conclusion

This study proposes a diagnostic-guided learning framework called SD-UNet. This architecture incorporates a coarse-to-refined clinical diagnostic strategy. SD-UNet incorporates two core innovations: First, DCDM utilizes an initial coarse mask prediction (ROI) to dynamically adjust the weights of small-kernel convolution feature aggregation, guiding subsequent edge refinement stages. This progressive refinement strategy allows the model to focus on relevant anatomical structures while effectively suppressing background noise. Second, STConv constructs a grouped structure tensor by computing local spatial derivatives, thereby obtaining physically meaningful directional prior information. Based on this directional prior, the model can drive the convolutional kernels to rotate dynamically, making them more closely fit the target contour. Comprehensive evaluations on five benchmark datasets show that SD-UNet typically achieves state-of-the-art performance. These results strongly demonstrate the robustness and generality of the proposed method. Future research will explore how to extend this architecture to improve instance-level discrimination against dense anatomical backgrounds and further optimize the STConv module to achieve a better balance between performance and processing efficiency.

CRedit authorship contribution statement

Jing Zhang: Conceptualization, Methodology, Writing - Original Draft, Writing - Review & Editing. **Erdong Yuan:** Conceptualization, Methodology, Writing - Original Draft, Writing - Review & Editing. **Zhaozhao Su:** Writing – review & editing, Visualization, Validation. **Gao Huang:** Writing – review & editing, Funding acquisition.

Yuxuan Luo: Software, Data curation, Validation. **Yining Yang:** Supervision, Project administration, Funding acquisition. **Zitong Guo:** Supervision, Project administration, Funding acquisition. **Jing Tao:** Supervision, Project administration, Funding acquisition. **Liejun Wang:** Supervision, Funding acquisition, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

I have shared the link (<https://github.com/Anonymous-Submission/SD-UNet>) to the dataset and code on GitHub, as mentioned in the abstract.

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